

Simanga Mchunu

Machine Learning Engineer | AI Researcher | Applied Mathematician | Backend Software Engineer
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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [ORCID](#) | [Sciforum Paper](#)

PROFESSIONAL SUMMARY

Machine Learning Engineer and AI Researcher with expertise in deep learning, probability theory, backend systems and mathematical modeling. Published researcher in automatic speech recognition for low-resource African languages and developer of Expanded Sample Space Theory (ESST). Experienced in PyTorch, FastAPI, cloud deployment and scalable AI systems. Passionate about trustworthy AI, computational mathematics and real-world machine learning applications.

EDUCATION

UNIVERSITY OF SOUTH AFRICA	Jan 2022 - Dec 2026
APPLIED MATHEMATICS AND COMPUTER SCIENCE BSC	Pretoria
Percentage: 70%	
ALX	July 2022 - Aug 2023
FULL-STACK SOFTWARE ENGINEERING-BACKEND CERTIFICATE	Johannesburg
Percentage: 90%	
UDACITY	March 2020 - October 2020
DATA SCIENCE CERTIFICATE	Online
Percentage: 80%	
A2SV	Jan 2026 - Dec 2026
DATA STRUCTURE AND ALGORITHMS CERTIFICATES	Ethiopia

RESEARCH EXPERIENCE

MACHINE LEARNING RESEARCHER Independent / UNISA Mathematical Sciences	2026 – Present
- Conduct research in probability theory, machine learning and mathematical modeling, with a focus on ASR for low-resource African languages. - Developed Expanded Sample Space Theory (ESST), a framework extending classical probability to real-world uncertainty. - Published an ASR paper on Sciforum and presented it as a contributed talk at Deep Learning IndabaX South Africa 2026; ESST accepted for oral presentation at ICAAD 2026 and as a poster at SASA 2026.	

EXPERIENCE

PYTORCH FOUNDATION AMBASSADOR Community Leadership	2026 – Present
- Selected as a 2026 PyTorch Foundation Ambassador. - Promote PyTorch through community engagement, technical education, workshops and mentorship. - Verify: PyTorch Ambassador Program – Application #363	
STATISTICS SOUTH AFRICA FIELD OFFICER ORGANIZER	Pretoria Dec 2021 – May 2022
- Publicity, Community Mobilisation and Advocacy - Census Geography - Data Collector	
RADIO1584 JUNIOR WEB DEVELOPER	Pretoria July 2015 – June 2018
- Assisting the web development team with all aspects of website and application design. - Assisting with the testing and maintenance of backend and front-end applications. - Collaborating with developers to implement new web features. - Keeping up-to-date with the latest technology and programming trends	
FRESHLY FARM INTERN: BACKEND ENGINEER TEAM LEADER	Online July 2024 – Jan 2025
Software development lifecycle, including architecture, solution design, quality assurance, maintenance and database management; provided technical expertise and leadership to the team.	

- Training students on Statistics, Algorithms and unsupervised Machine Learning
- Datathon and data science lifecycle
- Building projects

SHOOTING STAR FOUNDATION/IBM | MENTOR, IBM Z DATATHON 2025

Pretoria | October 2025

- Guided over 14,000 students from 44 countries in applying AI to real-world challenges under the theme Technology for Good. Supported participants with LinuxONE VM setup, provided technical and strategic mentorship, and developed leadership, communication and problem-solving skills through collaborative engagement.

PUBLICATIONS**Algorithm Design and Mathematical Modeling for Efficient Automatic Speech Recognition in Low-Resource African Languages**

Published, Sciforum (2026). Topics: ASR, mathematical modeling, low-resource African languages, machine learning.

Verify: sciforum.net/paper/view/31494**Expanded Sample Space Theory: A Comprehensive Framework for Real-World Probability**

Accepted for oral presentation, ICAAD 2026 (submission #648).

Verify: [Acceptance letter \(PDF\)](#)

CONFERENCE PRESENTATIONS

DEEP LEARNING INDABAX SOUTH AFRICA 2026 | CONTRIBUTED TALK

Durban | 8 July 2026

- Presented: Algorithm Design and Mathematical Modeling for Efficient Automatic Speech Recognition in Low-Resource African Languages (Simanga Comfort Mchunu). University of KwaZulu-Natal, Westville Campus; Theatre 2, 11:30.
- Listed in the official programme under Day 3 (Wednesday 8 July), Contributed Talks, Theatre 2.
- Verify: [Talk recording \(YouTube\)](#)

SASA 2026 CONFERENCE | POSTER PRESENTATION

Boksburg, South Africa

- Expanded Sample Space Theory (ESST): A Comprehensive Framework for Real-World Probability – Abstract ID A100, accepted by the Scientific Program Committee.
- Verify: sasaconference2026.org

ICAAD 2026 | ORAL PRESENTATION

Accepted

- Expanded Sample Space Theory: A Comprehensive Framework for Real-World Probability.

SKILLS**Programming Languages:** Python, C, C++, JavaScript, TypeScript, Assembly, R, Dart**Machine Learning / AI:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, Deep Learning, NLP, Computer Vision, LLMs, Reinforcement Learning, Speech Recognition, MLOps**Libraries/Frameworks:** React, Node.js, Angular, Django, Flask, FastAPI, RESTful APIs, Docker, Kubernetes, Nginx, Postman, Design Patterns, Data Analysis**Tools / Platforms:** Git, VS Code, Linux, Vim, Tableau, Power BI**Databases:** SQL, MongoDB, PostgreSQL

RESEARCH INTERESTS

Machine Learning, Artificial Intelligence, Deep Learning, Mathematical Modeling, Probability Theory, Speech Recognition, Large Language Models, Computational Mathematics, AI for African Languages

PROJECTS / OPEN-SOURCEE-COMMERCE BACKEND SYSTEM | [Live API](#)

Django, Swagger, Python, DRF, PostgreSQL

ALX Project Nexus – a professional e-commerce backend API built with Django REST Framework and PostgreSQL, demonstrating backend best practices and real-world application architecture.

AI-driven micro-bursary and peer-to-peer funding platform integrated with a student banking system. Students request support for fees, tuition or living costs; donors set preferences; AI matches students with donors on financial need, academic trajectory and donor intent; funds are disbursed instantly into student accounts.

E-commerce solution for South African spaza shops, bringing informal convenience stores onto digital retail channels.

Addresses language barriers in healthcare: an AI system performing speech-to-text, translation and text-to-speech, optimised through model fine-tuning and deployed on Google Cloud. I contributed to building it.

CERTIFICATIONS

- Introduction to Data Science and Computational Thinking – Stellenbosch University
- Introduction to Data Engineering – Coursera
- Quantum Computing and Programming Diploma – QWorld
- Mathematical Summer in Paris – University of Paris
- Mathematics Problem Solving – Limina
- Introduction to Statistics – Coursera
- AWS Machine Learning Foundation – Udacity
- Programming for Data Science – Udacity
- Data Analytics Fundamentals – IBM
- Introduction to SQL – DataCamp
- Introduction to Software Development – Power Learn Project (PLP)
- Kubernetes and Cloud Native Associate – The Linux Foundation, Andela
- ProDev Backend – ALX Africa

HONORS & AWARDS

- 2026 PyTorch Foundation Ambassador (Selected) – [verify](#)
- Oral Presentation Accepted, ICAAD 2026 – Expanded Sample Space Theory – [verify](#)
- SASA 2026 Conference – Poster Presentation (Accepted Abstract A100) – [verify](#)
- AOA Rising Stars Online Training 2026 – Excellence Certificate – [verify](#)
- Top Olympiad Mathematics Student – AOA Class of 2026
- Published Research: ASR for Low-Resource African Languages – [verify](#)
- Speaker (Contributed Talk): Deep Learning IndabaX South Africa 2026, UKZN Durban
- Top 15 Young Geek Award, Class of 2026
- Research/Publication: International Mathematics Conference
- Selected Top 20 SA Mathematician Camp
- Won ALX Hackathon 2024
- 2x Tableau Ambassador
- Finalist, DIRISA Datathon Challenge at CSIR
- Won IBM Z Datathon 2024 (Global winner number 1)
- 2nd Place, Zindi/IndabaX South Africa 2025 Hackathon with InstaDeep
- Zindi Ambassador 2025/26
- DIRISA Datathon Judge 2025
- Technical Mentor and winner, DIRISA 2025
- Datathon Technical Judge, Final Challenge 2025
- HackerRank Campus Crew Ambassador